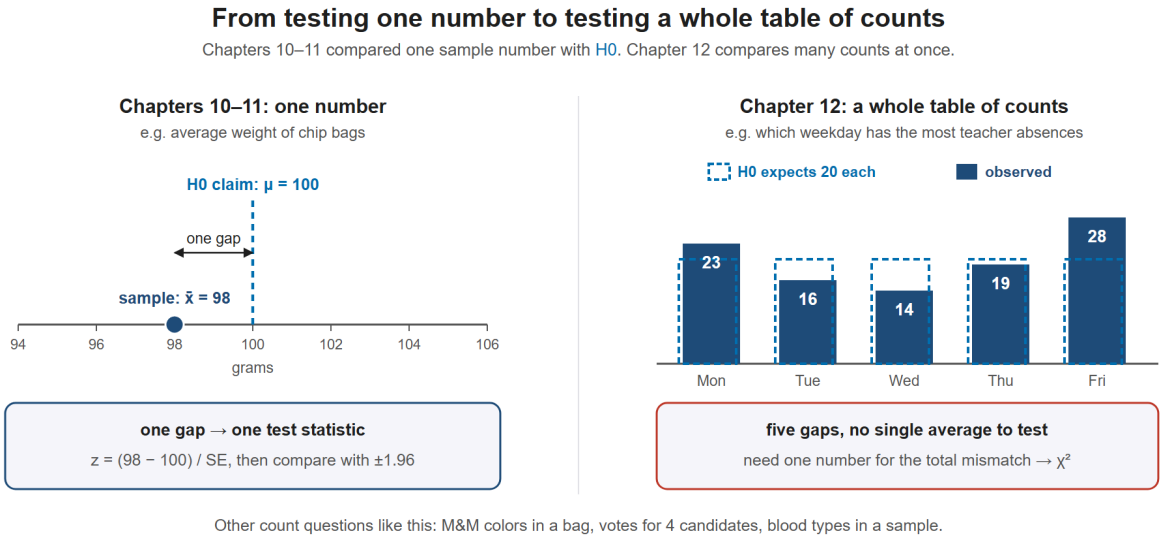
Lesson 12.1 - The Chi Square Test

> **Video:** Chi Square Test by The Organic Chemistry Tutor **Chapter:** 12 - The Chi-Square Distribution

The big picture: testing counts, not averages

Chapters 10 and 11 tested **one number**: a mean, a proportion, or a difference. Many questions are about **how counts are spread across several categories**: absences on each weekday, M&M colors in a bag, votes for 4 candidates. There is no single average to test. We need a way to compare a **whole table of counts** with what H_0 predicts.



Other count questions like this: M&M colors in a bag, votes for 4 candidates, blood types in a sample.

Left: one sample number and one claim give one gap, which becomes z or t. Right: a table of counts has a gap in every category. Chapter 12 combines all those gaps into one number, χ^2.

1. H_0 gives the expected counts. H_0 says what share each category should get. Multiply by the sample size:

$$E = n \times p$$

With 100 teachers and "absences are spread equally over 5 days", each day expects $E = 100 \times \frac{1}{5} = 20$.

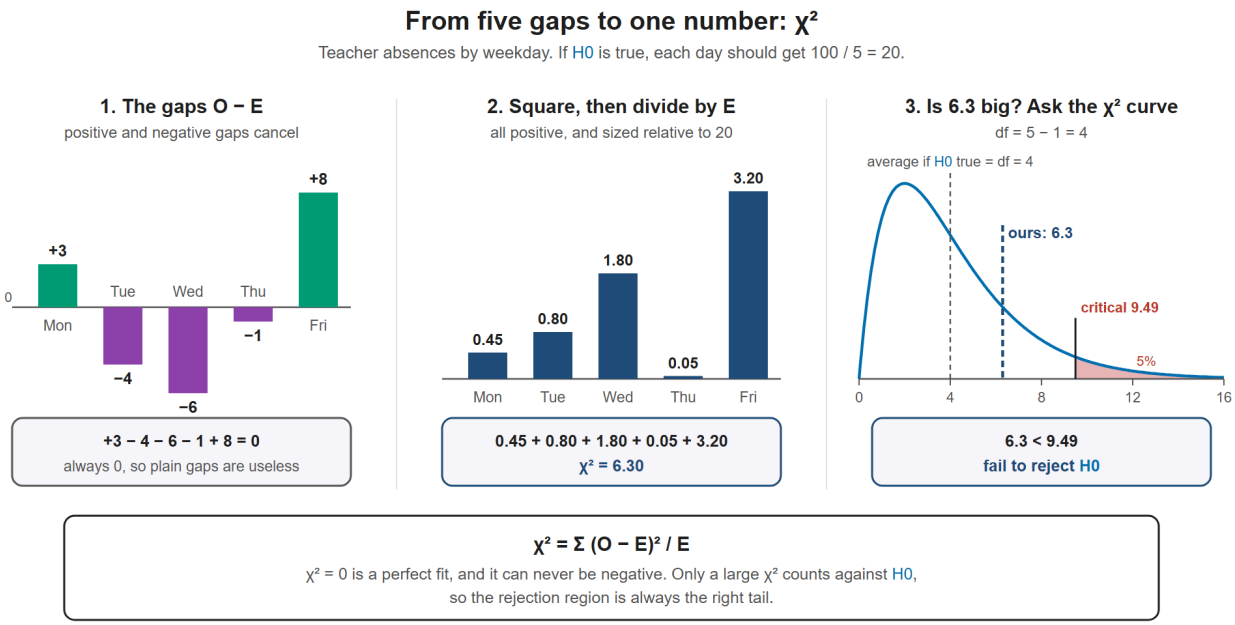
2. Why not just add up the gaps $O - E$? Because they always add up to 0. The observed and expected counts both total 100, so every extra absence on one day is a missing one somewhere else: $+3 - 4 - 6 - 1 + 8 = 0$.

3. Square, then divide by E.

- **Squaring** makes every gap positive, so the gaps can't cancel. It also gives big gaps extra weight.
- **Dividing by E** measures each gap **relative to what was expected**. Missing by 8 is a lot when you expect 20, but almost nothing when you expect 1000.

Add up the pieces for every category, and you get one number for the total mismatch:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$



The five gaps cancel out, so they are squared and scaled by E before being added. The total, 6.3, is then compared with the χ^2 curve to see whether it is unusually large.

4. Why the test is always right-tailed. χ^2 can't be negative, and $\chi^2 = 0$ means a perfect fit. Only a **large** χ^2 is evidence against H_0, so the rejection region is always the right tail. A χ^2 near 0 just means the data fit very well.

5. Where the χ^2 curve comes from. Each piece $\frac{O-E}{\sqrt{E}}$ behaves roughly like a z-score, so χ^2 is roughly a **sum of squared z-scores**. That sum has its own distribution, the **chi-square distribution**:

- It starts at 0 and is skewed right, because squares are never negative.
- Its shape depends on the **degrees of freedom**. On average, χ^2 is about **df** when H_0 is true. Here df = 4 and we got 6.3, a bit more than average but not unusual.

The chi-square distribution formula. If $Z_1, \ldots, Z_k$ are independent standard normal variables, then

$$\chi^2 = Z_1^2 + Z_2^2 + \cdots + Z_k^2$$

follows a chi-square distribution with k degrees of freedom. Its curve (probability density function) is

$$f(x) = \frac{x^{k/2-1}\, e^{-x/2}}{2^{k/2}\, \Gamma\left(\frac{k}{2}\right)}, \qquad x > 0$$

Quantity	Formula
Mean	$\mu = k$
Variance	$\sigma^2 = 2k$
Standard deviation	$\sigma = \sqrt{2k}$

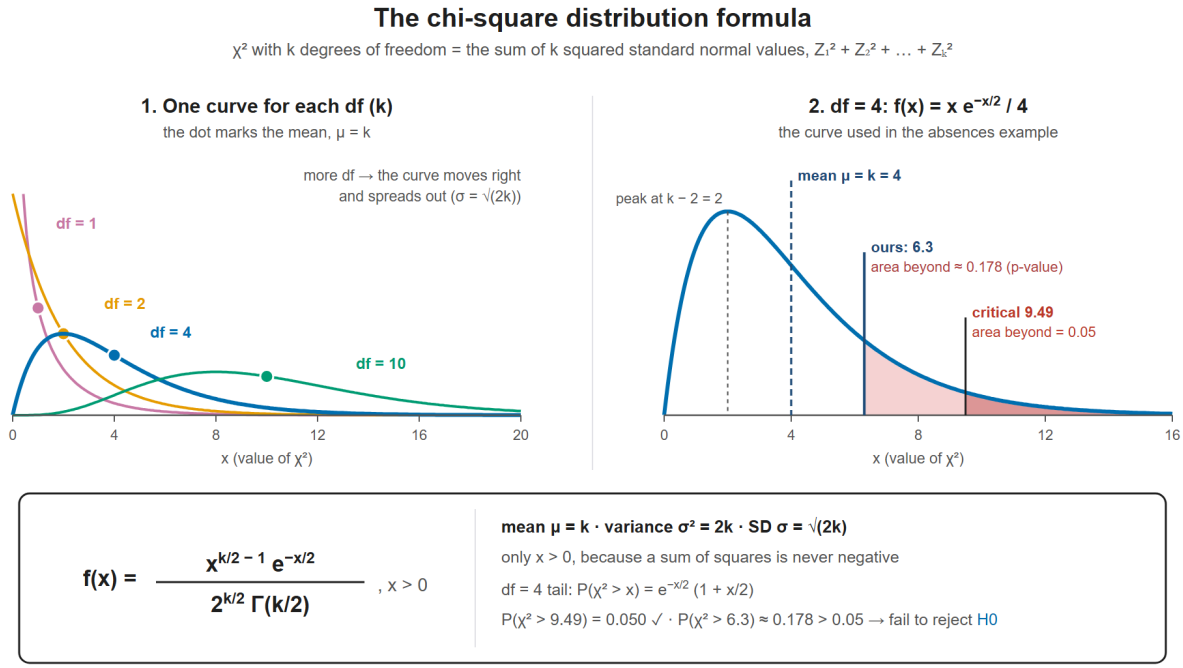
- k is the degrees of freedom (df).
- Γ is the gamma function, which extends the factorial: $\Gamma(n) = (n-1)!$ for whole numbers, and $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.
- $x > 0$ only, because a sum of squares can't be negative. That is why the curve starts at 0.
- The mean $\mu = k$ is why χ^2 "averages about df" when H_0 is true.

Example: df = 4. Here $\Gamma(2) = 1! = 1$ and $2^{4/2} = 4$, so the curve is

$$f(x) = \frac{x^{\,4/2-1}\, e^{-x/2}}{2^{4/2}\,\Gamma(2)} = \frac{x\, e^{-x/2}}{4}$$

For df = 4 the right-tail area also has a short closed form, $P(\chi^2 > x) = e^{-x/2}\left(1 + \frac{x}{2}\right)$. This checks the table and gives the p-value:

- At the critical value: $P(\chi^2 > 9.49) = e^{-4.745}(1 + 4.745) \approx 0.0087 \times 5.745 \approx 0.050$ ✓. That is exactly the 5% the table promises.
- At our value: $P(\chi^2 > 6.3) = e^{-3.15}(1 + 3.15) \approx 0.0429 \times 4.15 \approx 0.178$. The p-value is about 0.18, which is greater than 0.05, so we fail to reject H_0.



Left: each df has its own curve. The mean sits at k, so the curve slides right and spreads out as df grows. Right: the df = 4 curve from the absences example. The p-value (light red, about 0.178) is the area beyond our 6.3. The darker tail beyond 9.49 is the 5% rejection region.

You won't usually compute $f(x)$ by hand. The χ^2 table (or software) already lists the areas. The formula explains the curve's shape: it starts at 0, rises, then has a long right tail.

6. Why df $= k - 1$. The counts must add up to the total. Once you know 4 of the 5 days, the 5th is forced: $100 - 23 - 16 - 14 - 19 = 28$. Only $k - 1$ counts are free to vary.

The 5 steps are the same as in Chapters 10 and 11:

1. Write H_0 (the expected distribution) and H_a (it doesn't fit).
2. Choose α.
3. Find the critical value from the χ^2 table, with df $= k - 1$ and right-tail area α.
4. **Compute** $\chi^2 = \sum(O - E)^2/E$. ← this lesson
5. χ^2 greater than the critical value → reject H_0. Otherwise → fail to reject H_0.

The goodness-of-fit test

Does the **observed** distribution of counts across categories match an **expected** distribution?

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

- O = observed count, E = expected count, for each category
- **Degrees of freedom** = $k-1$, where k = number of categories
- It is always a **right-tailed** test: a large χ^2 means the observed counts are far from the expected ones

The chi-square distribution is **not symmetric**; it is skewed to the right. The critical value χ^2_α cuts off area α in the right tail:

- calculated χ^2 > critical value → **rejection region** → reject H_0
- calculated χ^2 ≤ critical value → **do-not-reject region**

Example - which weekday has the most absences?

A principal expects absences to be spread **equally** over the 5-day school week. 100 teachers were asked which day had the most absences. Test at $\alpha = 0.05$.

Day	Mon	Tue	Wed	Thu	Fri	Total
Observed O	23	16	14	19	28	100
Expected E	20	20	20	20	20	100

Expected counts: $E = n \times p = 100 \times \frac{1}{5} = 20$ for each day.

Hypotheses:

- H_0: absences happen with **equal frequencies** (a uniform distribution)
- H_a: absences do **not** happen with equal frequencies

Critical value: df $= k-1 = 5-1 = 4$. In the χ² table with right-tail area 0.05: $\chi^2_{0.05,\,4} = 9.49$.

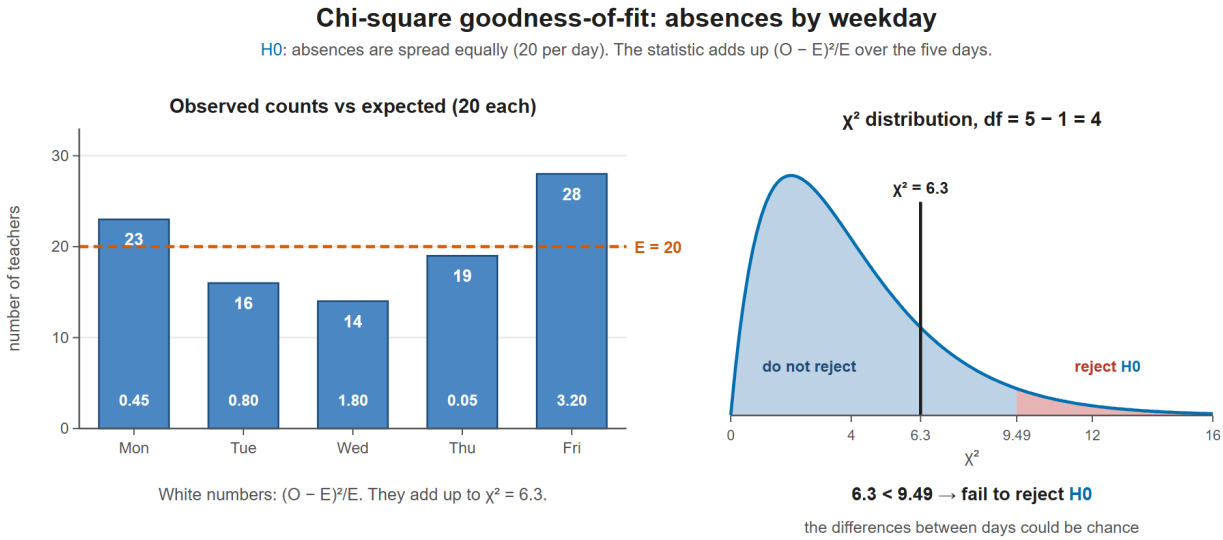
Test statistic:

Day	$O-E$	$(O-E)^2$	$(O-E)^2/E$
Mon	3	9	0.45
Tue	−4	16	0.80
Wed	−6	36	1.80
Thu	−1	1	0.05
Fri	8	64	3.20
Sum		**126**	**6.30**

$$\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{9+16+36+1+64}{20} = \frac{126}{20} = 6.3$$

(Every E is 20 here, so you can add the squares first and divide once.)

$6.3 < 9.49$, which is in the do-not-reject region. **Fail to reject H_0.** The variation between days is not large enough to say that absences are unevenly spread across the week.



Left: Friday and Wednesday are furthest from the expected 20, so they contribute the most to χ^2. Right: the total, 6.3, is still below the critical value 9.49, so the uneven pattern could be due to chance.

Key takeaways

- The gaps $O - E$ always add up to 0, so they are squared and divided by E before being added.
- $\chi^2 = \sum(O - E)^2/E$ measures how far the observed counts are from the expected ones.
- Goodness of fit: df = number of categories − 1. It is always right-tailed.
- Large χ^2 (beyond the critical value) → reject H_0 (the data don't fit the expected distribution).

Lesson 12.2 - The Chi Square Distribution Test of a Single Standard Deviation

> **Video:** Chi Square Distribution Test of a Single Variance or Standard Deviation by The Organic Chemistry Tutor **Chapter:** 12 - The Chi-Square Distribution

The big picture: testing spread instead of center

Every test so far has been about a **center**: a mean or a proportion. Sometimes the question is about **consistency**:

- Does a machine fill bags more evenly after a repair?
- Are this professor's exam scores less spread out than the department's?
- Is a new investment less risky (less variable)?

These questions make a claim about the population **standard deviation** σ (or the variance σ^2), so we need a test for spread.

1. Compare the sample's spread with the claim. The natural measure is the ratio

$$\frac{s^2}{\sigma_0^2}$$

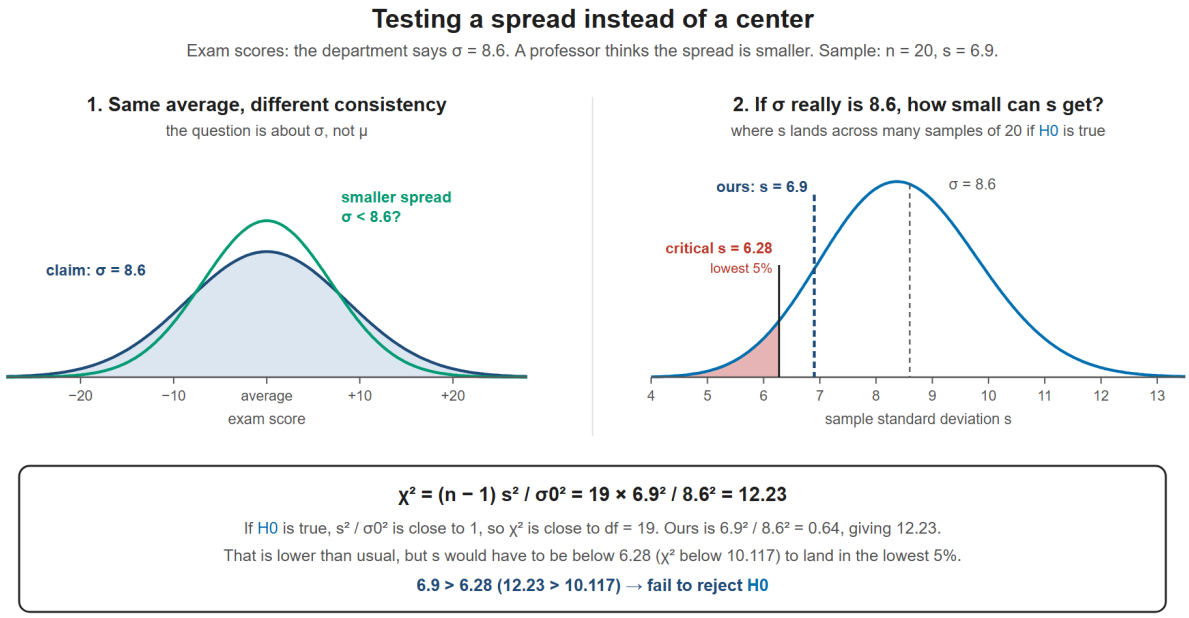
If H_0 is true, this ratio should be close to 1. A ratio well below 1 suggests a smaller spread, and well above 1 a bigger one. In the example, $6.9^2/8.6^2 = 0.64$. Is that far enough below 1?

2. Why multiply by $(n-1)$? The sample variance is $s^2 = \frac{\sum(x-\bar{x})^2}{n-1}$, so $(n-1)s^2 = \sum(x-\bar{x})^2$. Then:

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} = \sum\left(\frac{x-\bar{x}}{\sigma_0}\right)^2$$

That is a **sum of squared z-scores**, just like in Lesson 12.1. It follows the χ^2 distribution, provided the population is roughly normal.

3. Why df $= n-1$? The deviations $x-\bar{x}$ always add up to 0. Once you know $n-1$ of them, the last one is forced. This is the same reason that df $= k-1$ in Lesson 12.1. When H_0 is true, χ^2 averages about df. Here that's 19, and we got 12.23: lower than usual, but is it low enough?



Even when σ really is 8.6, samples of 20 give a range of s values: about 90% fall between 6.3 and 10.8. To land in the lowest 5%, s would have to be below 6.28 (that is, χ^2 below 10.117). Our $s = 6.9$ isn't low enough.

4. The curve is not symmetric. χ^2 can't be negative and is skewed right, so the left and right critical values are **not** mirror images like ± 1.96. χ^2 tables list **right-tail** areas:

- **Right-tailed test:** look up α.
- **Left-tailed test:** look up $1 - \alpha$ (here, 0.95).
- **Two-tailed test:** look up both $1 - \alpha/2$ and $\alpha/2$.

5. Variance or standard deviation, it's the same test. A claim about σ is a claim about σ^2. Square the claimed SD: $\sigma_0 = 8.6$ becomes $\sigma_0^2 = 73.96$.

The 5 steps are the same as always. Only step 4 uses a new formula:

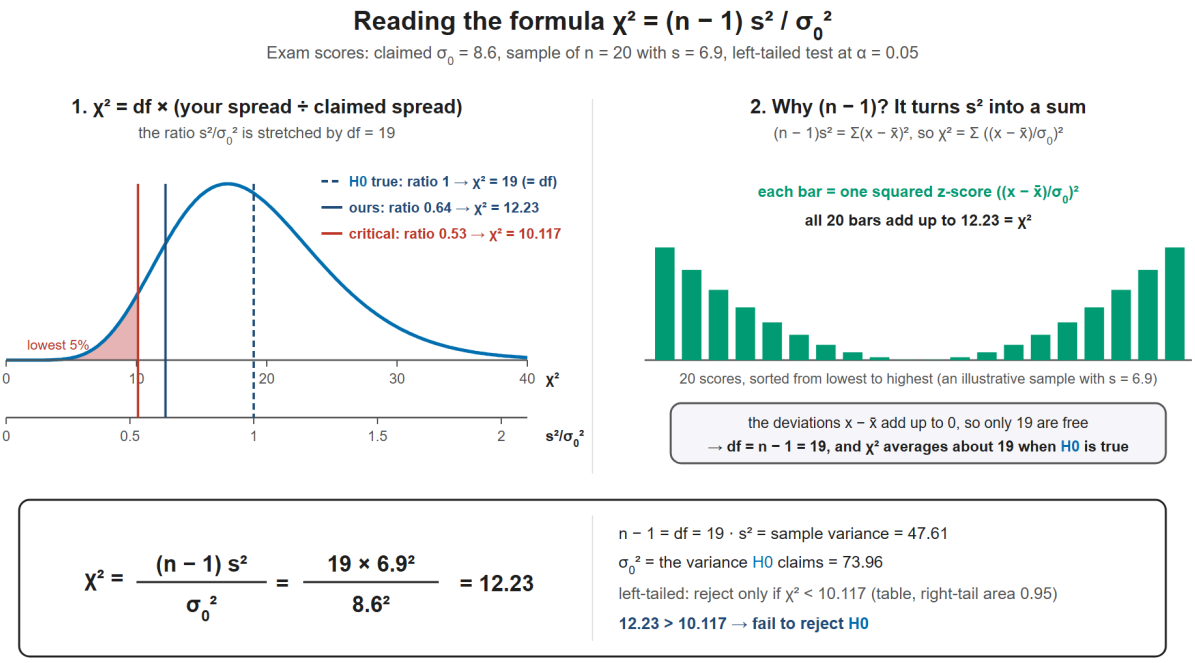
1. Write H_0 ($\sigma^2 = \sigma_0^2$) and H_a ($<$, $>$ or $\neq$).
2. Choose α.
3. Find the χ^2 critical value(s) with df $= n - 1$, using right-tail areas.
4. **Compute** $\chi^2 = (n-1)s^2/\sigma_0^2$. ← this lesson
5. In the rejection region → reject H_0. Otherwise → fail to reject H_0.

Test statistic

To test a claim about a population **variance** σ^2 (or standard deviation σ):

$$\chi^2 = \frac{(n-1)\,s^2}{\sigma_0^2}, \qquad \text{df} = n - 1$$

- s^2 = sample variance
- σ_0^2 = the variance claimed in H_0



Two ways to read the formula. Left: $\chi^2 = \text{df} \times \dfrac{s^2}{\sigma_0^2}$, the spread ratio stretched by df. A ratio of 1 (what H_0 expects) becomes χ^2 = df = 19. Our ratio of 0.64 becomes 12.23, which is not below the critical value 10.117 (ratio 0.53). Right: multiplying by $n-1$ turns s^2 back into a sum of squared deviations, so χ^2 is a sum of squared z-scores. That's why it follows the χ^2 distribution, with df $= n-1$.

H_a	Test	Rejection region
$\sigma^2 < \sigma_0^2$	Left-tailed	$\chi^2 <$ critical value, with area α on the left (look up right-tail area $1-\alpha$)
$\sigma^2 > \sigma_0^2$	Right-tailed	$\chi^2 >$ critical value, with right-tail area α
$\sigma^2 \neq \sigma_0^2$	Two-tailed	both tails, $\alpha/2$ each

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Example - is the spread of exam scores smaller?

The math department's exam scores have $\sigma = 8.6$. A professor thinks it is **less**. A sample of 20 scores has $s = 6.9$. Test at $\alpha = 0.05$.

Known values: $\sigma_0^2 = 8.6^2,\ \ s^2 = 6.9^2,\ \ n = 20,\ \ \text{df} = n - 1 = 19$

Hypotheses:

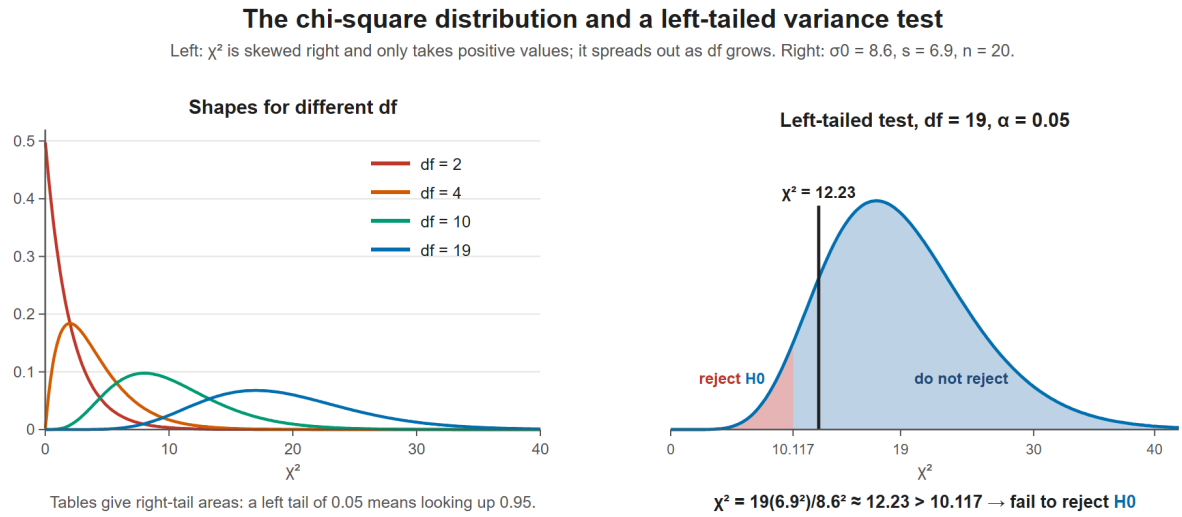
$$H_0 : \sigma^2 = 8.6^2 \qquad H_a : \sigma^2 < 8.6^2 \quad \text{(left-tailed)}$$

Critical value: the left-tail area is 0.05, so the right-tail area is 0.95. With df = 19 the table gives $\chi^2 = 10.117$. Reject H_0 only if the calculated χ^2 is **less than 10.117**.

Test statistic:

$$\chi^2 = \frac{(n-1)\,s^2}{\sigma_0^2} = \frac{19 \times 6.9^2}{8.6^2} = \frac{19 \times 47.61}{73.96} \approx 12.23$$

$12.23 > 10.117$, which is in the do-not-reject region. **Fail to reject H_0.** At the 5% level the sample does not provide enough evidence that the standard deviation is less than 8.6.



Left: χ^2 is never negative and is skewed to the right; with more degrees of freedom it moves right and spreads out. Right: for the left-tailed variance test the rejection region is the left tail below 10.117, and 12.23 does not reach it.

Key takeaways

- Testing a variance or SD: $\chi^2 = (n-1)s^2/\sigma_0^2$ with df $= n - 1$.
- If H_0 is true, $s^2/\sigma_0^2 \approx 1$, so χ^2 is near df. A χ^2 far below df suggests less spread, and far above suggests more.
- χ^2 tables list **right-tail areas**. For a left-tailed test, look up $1 - \alpha$.
- Left-tailed: reject only when χ^2 is **below** the critical value.

Lesson 12.3 - Tests of Independence Using Chi-Square Distribution

> **Video:** Test of Independence Using Chi-Square Distribution by The Organic Chemistry Tutor **Chapter:** 12 - The Chi-Square Distribution

The big picture: are two categories related?

Lesson 12.1 compared **one** categorical variable with a distribution that H_0 stated in advance. Here there are **two** categorical variables measured on the same people, for example student type (freshman or senior) and daily study time (0-2, 2-4 or 4-6 hours). The question: **does knowing one tell you anything about the other?**

1. Count the combinations in a table. Each student falls into one row and one column. The result is a **contingency table** of counts, with row totals, column totals and a grand total.

2. What would "no relationship" look like? If study time had nothing to do with student type, freshmen and seniors would split across the study-time columns in the **same proportions**: the split of all 630 students. Overall, $223/630 = 35.4\%$ study 0-2 hours, so we'd expect 35.4% of the 310 freshmen to do so too:

$$E = 310 \times \frac{223}{630} = 109.7$$

3. Where $E = \dfrac{\text{row total} \times \text{column total}}{\text{grand total}}$ comes from. Use the multiplication rule for independent events (Lesson 5.5):

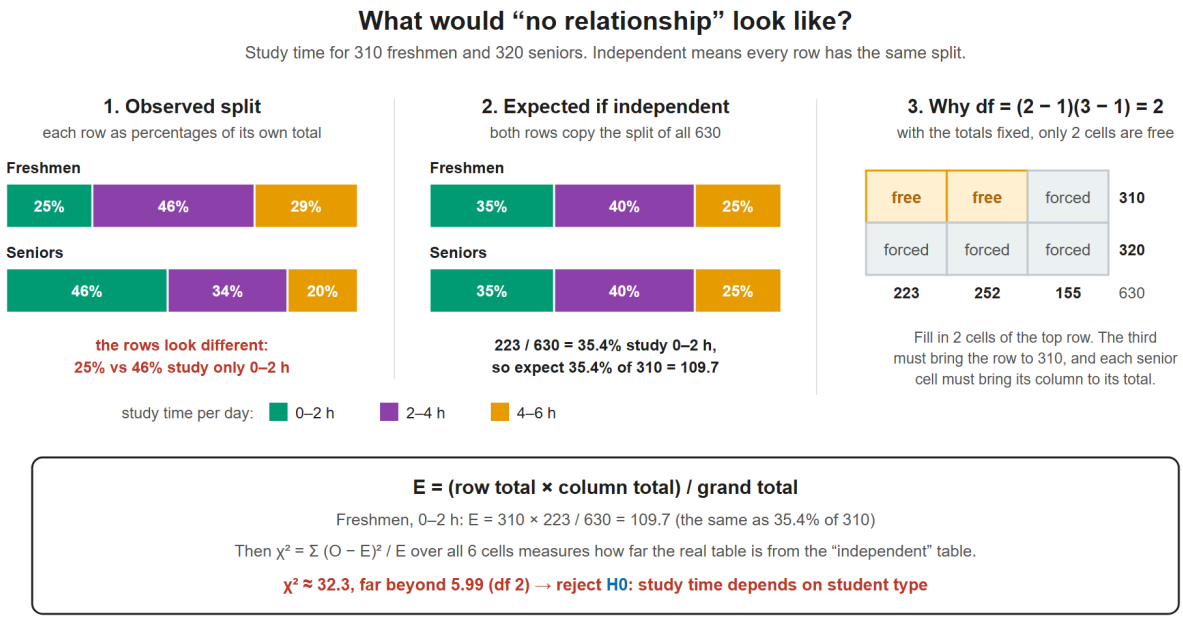
$$P(\text{row and column}) = P(\text{row}) \times P(\text{column}) = \frac{\text{row total}}{N} \times \frac{\text{column total}}{N}$$

To turn a probability into a count, multiply by N:

$$E = N \times \frac{\text{row total}}{N} \times \frac{\text{column total}}{N} = \frac{\text{row total} \times \text{column total}}{N}$$

So H_0 (independence) gives us every expected count, built from the table's own totals.

4. Then it's the same χ^2 as Lesson 12.1. Compare each observed count with its expected count, and add up $\frac{(O-E)^2}{E}$ over **every cell** of the table. A large χ^2 means the real table is far from the "independent" table.



Left: in the real data the freshman and senior rows look different. Middle: under independence both rows would have the same split. χ^2 measures the distance between these two pictures.

5. Why df $= (r-1)(c-1)$. The row and column totals are fixed. In each row, the last cell is forced by the row total, which leaves $c-1$ free cells per row. The last row is forced by the column totals, which leaves $r-1$ free rows. Here that is $(2-1)(3-1) = 2$ free cells.

6. Related, not "causes". Rejecting H_0 says the variables are **associated**. It doesn't say why. Seniors might study less for many reasons, and the test can't tell which.

How this compares with Lesson 12.1:

	Goodness of fit (12.1)	Independence (this lesson)
Variables	one	two
Expected counts come from	the proportions that H_0 states	the table's own row and column totals
df	$k-1$	$(r-1)(c-1)$
Test statistic	$\sum(O-E)^2/E$	$\sum(O-E)^2/E$ over all cells

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The idea

A **test of independence** checks whether two categorical variables in a **contingency table** are related.

	Formula
H_0	the variables are **independent**
H_a	the variables are **dependent** (related)
Expected count for each cell	$E = \dfrac{\text{row total} \times \text{column total}}{\text{grand total}}$
Test statistic	$\chi^2 = \sum \dfrac{(O-E)^2}{E}$ over **all** cells
Degrees of freedom	$(r-1)(c-1)$, where r = rows and c = columns

It is a **right-tailed** test: reject H_0 if χ^2 exceeds the critical value.

Example - study time vs type of student

Is the number of hours spent studying each day related to whether a student is a freshman or a senior? $\alpha = 0.05$.

Step 1 - complete the observed table

	0-2 h	2-4 h	4-6 h	**Row total**
Freshmen	76	143	91	**310**
Seniors	147	109	64	**320**
Column total	**223**	**252**	**155**	**630**

Step 2 - hypotheses and critical value

- H_0: study hours are **independent** of student type
- H_a: study hours **depend** on student type
- df $= (r-1)(c-1) = (2-1)(3-1) = 2$, right-tail area 0.05: $\chi^2_{\text{crit}} = 5.99$

Step 3 - expected counts

$$E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$

Example: $E_{1,1} = \dfrac{310 \times 223}{630} \approx 109.7$. The other cells use their own row and column totals, for example $E_{2,3} = \dfrac{320 \times 155}{630} \approx 78.7$.

	0-2 h	2-4 h	4-6 h
Freshmen	109.7 (≈110)	124.0	76.3 (≈76)
Seniors	113.3 (≈113)	128.0	78.7 (≈79)

The rounded values still match every row and column total (e.g. 110 + 124 + 76 = 310).

Step 4 - test statistic (with the video's rounded expected counts)

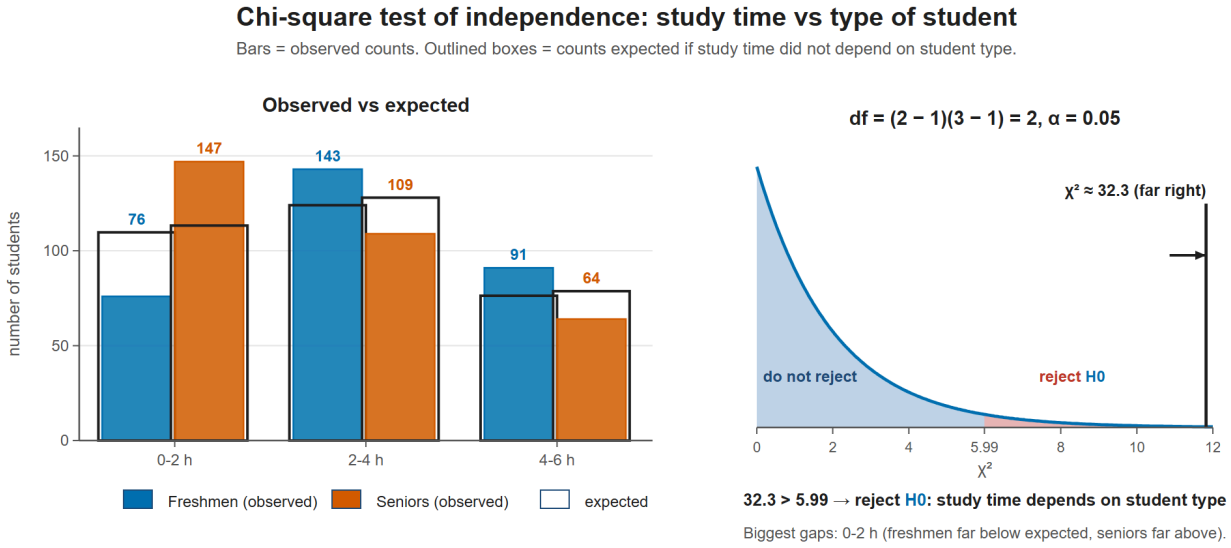
$$\chi^2 = \sum_{\text{all cells}} \frac{(O-E)^2}{E}, \qquad \text{for example, the first cell: } \frac{(76-110)^2}{110} = \frac{1156}{110} = 10.51$$

Cell	O	E	$(O-E)^2/E$
Freshmen, 0-2	76	110	10.51
Freshmen, 2-4	143	124	2.91
Freshmen, 4-6	91	76	2.96
Seniors, 0-2	147	113	10.23
Seniors, 2-4	109	128	2.82
Seniors, 4-6	64	79	2.85
Total			**≈ 32.3**

> **Note:** expected counts don't have to be whole numbers. Using the unrounded values (109.73, 124.0, 76.27, 113.27, 128.0, 78.73) gives $\chi^2 \approx 31.7$. The conclusion is the same.

Step 5 - decision

$32.3 > 5.99$, which is in the rejection region. **Reject H_0.** There is strong evidence that study time **depends** on student type. In the data, seniors are much more likely than freshmen to study only 0-2 hours a day.



Left: if study time were independent of student type, the bars would roughly fill the outlined expected boxes. The 0-2 hour group is far off for both freshmen and seniors. Right: $\chi^2 \approx 32.3$ is far beyond 5.99.

Key takeaways

- Under independence every row has the same split as the whole table. That is where expected count = (row total × column total) ÷ grand total comes from.
- $\chi^2 = \sum(O-E)^2/E$ over every cell, with df $= (r-1)(c-1)$.
- Right-tailed: a large χ^2 → reject independence → the variables are related.